SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Coding & Solution

Date	03JULY 2026
Team ID	
Project Name	***AComprehensive Measure of Well-Being***
Maximum Marks	5Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	***Details***
Repository Link / URL	https://github.com/Adithya1171/A-Comprehensive-Measure-of-Well-Being
Programming Language(s)	Python, HTML5, CSS3
Framework(s) Used	Flask, Scikit-learn
Key Features Implemented	HDI score prediction using four development indicators; HDI category classification; feature preprocessing and scaling; exploratory data analysis with 9 visualizations; saved ML model and scaler; interactive Flask web interface; prediction result display.
Pending / Incomplete Features	None – All core project requirements have been implemented and tested successfully.
Setup / Run Instructions	1. Open the HDI_Predictor folder in VS Code. 2. Activate the virtual environment. 3. Install dependencies using pip install -r requirements.txt. 4. Run python app.py. 5. Open the local URL displayed in the terminal. 6. Enter the four HDI parameters and click Predict.

CodeQualityChecklist

S.No	*Criteria*	*Status (Yes / No)*
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	No

Additional Notes / Comments:

The HDI Predictor has been successfully developed and tested in Visual Studio Code .The application follows a modular project structure with separate components for model training , preprocessing ,prediction logic, templates, static files, data, and saved model artifacts. The system accepts Life Expectancy ,Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita as inputs and generates an HDI prediction through a Flask web interface. All core project requirements have been completed successfully. Version control repository submission is not required for this project.